Lecture 6: How Do We Communicate?: Language in the Brain, Mouth and the Hands (full transcript)

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Overview: One of the most uniquely human abilities is the capacity for creating and understanding language. This lecture introduces students to the major topics within the study of language: phonology, morphology, syntax and recursion. This lecture also describes theories of language acquisition, arguments for the specialization of language, and the commonalities observed in different languages across cultures.

Chapter 1. The Scientific Notion of Language and Structure [00:00:00]

Professor Paul Bloom: This class today is about language. And language is, to a large extent, where the action is. The study of human language has been the battleground over different theories of human nature. So, every philosopher or psychologist or humanist or neuroscientist who has ever thought about people has had to make some claim about the nature of language and how it works. I'm including here people like Aristotle and Plato, Hume, Locke, Freud and Skinner. I'm also including modern-day approaches to computational theory, cognitive neuroscience, evolutionary theory and cultural psychology. If you hope to make it with a theory of what people are and how people work, you have to explain and talk about language. In fact, language is sufficiently interesting that, unlike most other things I'll talk about in this class, there is an entire field devoted to its study, the field of linguistics that is entirely devoted to studying the nuances and structures of different languages.

Now, I'll first, before getting into details, make a definitional point. When I'm talking about language I'm meaning systems like English and Dutch and Warlpiri and Italian and Turkish and Urdu and what we've seen and heard right now in class in the demonstration that preceded the formal lecture. [Before class started, Professor Bloom had several bilingual students give demonstrations of non-English speech.] Now, you could use language in a different sense. You could use the term "language" to describe what dogs do, or what chimpanzees do, or birds. You could use language to describe music, talk about the — a musical language or art, or any communicative system, and there's actually nothing wrong with that. There's no rule about how you're supposed to use the word "language." But the problem is if you use the word "language" impossibly, incredibly broadly, then from a scientific point of view it becomes useless to ask interesting questions about it. If language can refer to just about everything from English to traffic signals, then we're not going to be able to find interesting generalizations or do good science about it.

So, what I want to do is, I want to discuss the scientific notion of language, at first restricting myself to systems like English and Dutch and American sign language and Navajo and so on. Once we've made some generalizations about language in this narrow sense, we could then ask, and we will ask, to what extent do other systems such as animal communication systems relate to this narrower definition. So we could ask, in this narrow sense, what properties do languages have and then go on to ask, in a broader sense, what other communicative systems also possess those properties.

Well, some things are obvious about language so here are some; here are the questions we will ask. This will frame our discussion today. We'll first go over some basic facts about language. We'll talk about what languages share, we'll talk about how language develops, and we'll talk about language and communication in nonhumans.

I began this class with a demonstration of — that illustrates two very important facts about language. One is that languages all share some deep and intricate universals. In particular, all languages, at minimum, are powerful enough to convey an abstract notion like this; abstract in the sense that it talks about thoughts and it talks about a proposition and spatial relations in objects. There's no language in the world that you just cannot talk about abstract things with. Every language can do this. But the demonstration [before class] also illustrated another fact about language, which is how different languages are. They sound different. If you know one language, you don't necessarily know another. It's not merely that you can't understand it. It could sound strange or look unusual in the case of a sign language. And so, any adequate theory of language has to allow for both the commonalities and the differences across languages. And this is the puzzle faced by the psychology and cognitive science of language.

Well, let's start with an interesting claim about language made by Charles Darwin. So, Darwin writes, "Man has an instinctive tendency to speak, as we see in the babble of our young children, while no child has an instinctive tendency to bake, brew or write." And what Darwin is claiming here, and it's a controversial and interesting claim, is that language is special in that there's some sort of propensity or capacity or instinct for language unlike the other examples he gives. Not everything comes natural to us but Darwin suggests that language does.

Well, why should we believe this? Well, there are some basic facts that support Darwin's claim. For one thing, every normal — every human society has language. In the course of traveling, cultures encounter other cultures and they often encounter cultures that are very different from their own. But through the course of human history, nobody has ever encountered another group of humans that did not have a language. Does this show that it's built in? Well, not necessarily. It could be a cultural innovation. It could be, for instance, that language is such a good idea that every culture comes across it and develops it. Just about every culture uses some sort of utensils to eat food with, a knife and a fork, chopsticks, a spoon. This probably is not because use of eating utensils is human nature, but rather, it's because it's just a very useful thing that cultures discover over and over again. Well, we know that this probably is not true with regard to language. And one reason we know this is because of the demonstrated case studies where a language is created within a single generation. And these case studies have happened over history.

The standard example is people involved in the slave trade. The slave trade revolving around tobacco or cotton or coffee or sugar would tend to mix slaves and laborers from different language backgrounds, in part deliberately, so as to avoid the possibility of revolt. What would happen is these people who were enslaved from different cultures would develop a makeshift communication system so they could talk to one another. And this is called a "pidgin," p-i-d-g-i-n, a pidgin. And this pidgin was how they would talk. And this pidgin was not a language. It was strings of words borrowed from the different languages around them and put together in sort of haphazard ways.

The question is what happens to the children who are raised in this society. And you might expect it that they would come to speak a pidgin, but they don't. What happens is, in the course of a single generation, they develop their own language. They create a language with rich syntax and morphology and phonology, terms that we'll understand in a few minutes. And this language that they create is called a "creole." And languages that we know now as creoles, the word refers back to their history. That means that they were developed from pidgins. And this is interesting because this suggests that to some extent the ability to use and understand and learn language is part of human nature. It doesn't require an extensive cultural history. Rather, just about any normal child, even when not exposed to a full-fledged language, can create a language.

And more recently, there's been case studies of children who acquire sign language. There's a wonderful case in Nicaragua in sign language where they acquire sign language from adults who themselves are not versed in sign language. They're sort of second-language learners struggling along. What you might have expected would be the children would then use whatever system their adults use, but they don't. They "creolized" it. They take this makeshift communication system developed by adults and, again, they turn it into a full-blown language, suggesting that to some extent it's part of our human nature to create languages.

Also, every normal human has language. Not everybody in this room can ride a bicycle. Not everybody in this room can play chess. But everybody possesses at least one language. And everybody started to possess at least one language when they were a child. There are exceptions, but the exceptions come about due to some sort of brain damage. Any neurologically normal human will come to possess a language.

What else do we know? Well, the claim that language is part of human nature is supported by neurological studies, some of which were referred to in the chapters on the brain that you read earlier that talk about dedicated parts of the brain that work for language. And if parts of these brains — if parts — if these parts of the brain are damaged you get language deficits or aphasias where you might lose the ability to understand or create language. More speculatively, there has been some fairly recent work studying the genetic basis of language, looking at the genes that are directly responsible for the capacity to learn and use language. And one bit of evidence that these genes are implicated is that some unfortunate people have point mutations in these genes. And such people are unable to learn and use language.

So, in general, there is some support, at least at a very broad level, for the claim that language is in some sense part of human nature. Well, what do we mean by language? What are we talking about when we talk about language? We don't want to restrict ourselves, for instance, to English or French. What do all languages share? Well, all languages are creative and this means a couple of things.

One meaning is the meaning emphasized by Rene Descartes. When Rene Descartes argued that we are more than merely machines, his best piece of evidence for him was the human capacity for language. No machine could do this because our capacity for language is unbounded and free. We could say anything we choose to say. We have free will. And in fact, language allows us to produce a virtual infinity of sentences. So, we could create and understand sentences that we never heard before. And there are a lot of sentences. So, if you want to estimate how many grammatical sentences under twenty words in English, the answer is, "a lot." And what this means is that any theory of language use and

language comprehension cannot simply appeal to a list. When you understand a sentence I said you have to have the capacity to understand a sentence even if you've never heard it before. And this is because we could effortlessly produce and understand sentences that no human has ever said before on earth.

Would anybody volunteer to say a sentence, non obscene, non derogatory, that has never been spoken before on earth, ever? Here. I'll start. "It's surprisingly easy to get a purple tie on eBay if you don't care much about quality." I could imagine no one else in the world has said this before. "I am upset that one cannot easily download 'Buffy the Vampire Slayer' through iTunes." Now, it's possible somebody said both these sentences before, but you probably have not heard them. But you understand them immediately. So, how do you do it? Well, you have rules in your head. You've learnt what the words mean, but you have abstract and unconscious rules that take these words, figure out the order, and in a fraction of a second, give rise to understanding. And that's the sort of thing linguists study.

So, take some standard examples from the linguistic study of English. And bear in mind the rules we're talking about here are not rules you explicitly know. They're automatic rules of the same sort we're going to talk about in the context of visual perception in that they're implicit and unconscious and not accessible to explicit understanding. So for instance, immediately you read "The pig is eager to eat" versus "The pig is easy to eat" and in a fraction of a second you know there's an important difference. "The pig is eager to eat" means the state of affairs that we're talking about is when the pig does the eating. "The pig is easy to eat" is when the pig is being eaten.

You would see a sentence like "Bill knew that John liked him" and you know, without even knowing how you know, that this could mean that Bill knew that John liked Bill or it could mean that Bill knew that John liked Fred. But it can't mean that Bill knew that John liked John. The natural interpretation, in fact, is that Bill knew that John liked Bill. The two words co-refer. Contrast that with "Bill knew that John liked himself," which only has the meaning Bill knew that John liked John. And this is what linguists do for a living so if you hear me talking about this and say, "I want to spend the next forty years of my life studying that," you should become a linguist. But that's the sort of — those are the sort of phenomena that we're interested in.

Now, it gets more complicated. Those are examples from syntax, but language has many structures. Language has structures going from the bottom to the top. All languages — All human languages have phonology, which is the system of sounds or signs; morphology, which is the system of words or morphemes, basic units of meaning; and syntax, which refer to rules and principles that put together words and phrases into meaningful utterances. And I want to talk briefly about each of these three parts of language before looking at some other issues. I'm indebted here to Steven Pinker's excellent book The Language Instinct which provides, I think, a superb discussion of these phenomena. And I'm going to steal some of my examples from Pinker.

Chapter 2. Phonology: A System of Sounds [00:15:53]

So, phonology. Phonology is the system of sounds that languages have. There's a subset. There's a list, a finite list, of possible sounds that language can use. I'm going to put aside for the moment the question of sign languages and how they work. I'm going to talk about them in a little bit. The idea is that English has about forty of these phonemes. So, if you're a native monolingual speaker of English

you hear speech and each sound you hear is categorized as falling into one of those forty morphemes — sorry, phonemes. So, for example, English has a phoneme of "lu," "l," and a phoneme of "r." And so, an English speaker can hear the difference between "lip" and "rip" and that corresponds to two different words in English. Other languages don't have that distinction and so those distinctions are very difficult for non-native English speakers to learn.

So, part of what goes on when you learn, is you have to learn the language — the phonemes that your language has. Another part of the problem of learning language is you have to figure out what the boundaries are between the words. You have to use sound signals to figure out the boundaries between the words. Now that — If the only language you've ever heard is English, that's going to seem like a really weird example of a problem because you're listening to me speak and in between each of my words you're hearing a pause. You don't have to be very smart to figure out where one word begins and one word ends. But the pause is a psychological illusion. If you were to just talk into an oscilloscope that measured your sound vibrations, there are no pauses between the words. Rather, the pauses are inserted by your mind as you already know where one word begins and another one ends. And you insert a pause at that point.

You could see this when you hear a language you don't already know. So, for those of you who have never heard French before, when you hear somebody say, "Je ne sais pas" you could say, "Remarkable! French has no pauses between words." And you — And now a French speaker, of course, hears "Je ne sais pas." For Hebrew, I know one sentence in Hebrew: "Sleecha, eypho ha-sheeruteem" which I think is a request for the bathroom. But if you don't know Hebrew there's no pauses. And the truth is, when you each gave your demonstrations, nobody spoke properly because nobody spoke — Here's the sentence: "Glorp [pause] fendel [pause] smug [pause] wuggle." Rather, you all sounded like, "blublublublublub" without any pauses because I don't know your languages.

Children come into the world without knowing any specific language and so they have to learn pauses. They have to learn to interpret sounds in context and sometimes they make mistakes. They get problems of segmentation. And there are some illustrations. You could see their mistakes if they're trying to repeat back something that's already known within a society. So, songs are a good example. These are excerpts from children. [misunderstood lyrics from songs] "I'll never be your pizza burnin'." Anybody know — figure out what that corresponds to?

Student: [I'll never be your] Beast of burden?

Professor Paul Bloom: "Beast of burden." Very good. [reading another misunderstood song lyric off of the slide] "A girl with colitis goes by." Somebody?

Student: "A girl with kaleidoscope eyes."

Professor Paul Bloom: "The ants are my friends; they're blowin' in the wind." And [laughter] this is a religious one. "Our father with Bart in heaven; Harold be they name... Lead us not into Penn Station..."

Now, phonological understanding illustrates all sorts of aspects of language processing and, in fact, of consciousness. Because remember I said that, typically, when you hear a sentence you make — you manufacture in your mind gaps between the words. Typically, when there's something which is unclear you'll fill in the gap and figure out what the word is. And you'll hear it that way. So, the few examples — The best examples, again, are for when it goes wrong.

So, a classic example is from the song "Super Freak" by Rick James. I got a big lecture about copyright laws and this is going to violate most of them. Rick James is going to be sitting on the — at — staring at the web two years from now saying, "Hey. That's my thing." Okay. So, I want you to listen to this line. I'm sure most of you have heard this before but I want you to listen closely. [music playing] What was that last line? [laughter] "The kind of girl you read about — " Well, it turns out that nobody really knows. And it sounds to many people who do top-down interpretation as — to me as well, that "she's the kind of girl you read about in Newsweek magazine." But that makes no sense at all given that you don't want to "bring home to Mama." [referring to a song lyric] And she's — and it's not the — and in fact, if you check the notes on the song, she's in fact, "the kind of girl you read about in new wave magazines." Now, when you listen to it then, again, knowing that, you hear it that way. [music playing]

Now, this top-down — This is known as "top-down" processing. Top-down processing is an example of when you know what something is you hear it that way. And this is extremely useful when it comes to filling in gaps in sounds. In normal conversation, if I'm to say "s — [coughs] entence" you won't hear that as "s — [cough] entence." Rather, you hear "sentence." You fill in the gap. This can lead to problems. The problem it's led to in my life revolves around the song "Get Crunk" [laughter] because I've heard "Get Crunk" and my children asked me if I would buy them "Get Crunk" from iTunes. My children are eight and ten. And now "Get Crunk," as I was aware from having heard it before, involves the consistent refrain of "get crunk" extremely bad word, "get crunk" extremely bad word, and so I said "no." And then they said, "Well, there's a clean version of it." So, I downloaded the clean version. Unfortunately, knowing what the clean version — knowing what the word is means to me the clean version is not very clean. Now, I will add, [laughter] before people write letters and stuff, this is the clean version. [The music plays, but even though the expletives are censored out you still perceive them as being there] [laughter] Thank goodness they took away that obscene word. [laughter]

Okay. So, top-down processing affects how we hear things, usually, almost always, for the better. And in fact, this is a theme we're going to return to next class when we talk about vision because the same thing is going to happen there. How we see the world is often confusing and befuddled but what we know can clear things up. Same with sound.

Chapter 3. Morphology: A System of Words [00:24:07]

Morphology is the next level up. Phonology is sounds. Morphology is words. And human language uses this amazing trick described by Ferdinand de Saussure, the great linguist, as "the arbitrariness of the sign." And what this means is we can use — take any arbitrary idea in the world, the idea of a chair or a story or a country, and make a sound or a sign to connect to it. And the link is arbitrary. You might choose to use a word for "dog" as "woof woof" because it sounds like a dog but you can't use a word for "country" that sounds like a country. You could use a sign language thing for "drink" that looks sort of like the act of drinking but you can't use a sign language word for "country" that looks like a country, or for "idea" that looks like an idea.

So, the way languages work is it allows for arbitrary naming. It allows for this map between a symbol, say a spoken word, and any sort of thought we want to use. And those arbitrary mappings, as we come to learn them, make up the vocabulary of a language. I'm talking about words but the more technical term is "morpheme." And what a morpheme is is the smallest meaningful unit in a language. Now

often, this is the same thing as a word. So, "dog" is a word. And "dog" is also a morpheme, but not always because there are single morphemes and then there are words that are composed of many morphemes. So, "dogs" and "complained" are one word, but two morphemes and what this means is that you make the word by putting together two morphemes. To put it differently, in order to know what "dogs" means, you never had to learn the word "dogs." All you had to know is the word "dog" and the plural morpheme 's' and you could put them together to create a word.

How many morphemes does the average speaker know? The answer is fairly startling. The average speaker knows, as a low-ball estimate, about 60,000 words. I think the proper estimate is closer to 80,000 or 100,000. What this means, if you average it out, is that since children start learning their first words at about their first year of life, they learn about nine new words a day. And it's not a continuous nine words every day. It goes up and down depending on the age. But still, the amount of words we know is staggering. How many of you know more than one language pretty fluently? Those of you who know other languages might have in your heads 200,000 words or 300,000 words and you're accessing them in a fraction of a second. It is — could legitimately be seen as one of the most astonishing things that people do.

Chapter 4. Syntax: Communicating Complicated Ideas [00:27:21]

Finally, syntax. So, we have the sound system of a language, the phonology. We have the words of a language, the morphology, but all that gives you is "dog," "cup," "chair," "house," "story," "idea." That won't allow us to communicate complicated ideas. So, the final step in the story is syntax. And syntax refers to those rules and principles that allow us to combine words into phrases and phrases into sentences. And syntax uses another neat trick and this is defined by Wilhelm von Humboldt as the "infinite use of finite media." So, here's the question. Your vocabulary is finite. There are just so many words. You have to learn them one by one, but you could produce a virtual infinity of sentences. How can you do that? How can you go from a finite list of symbols to an infinite number of sentences? And the answer is you have a combinatorial system.

Now, language is not the only thing in culture or nature that has this sort of combinatorial system. Music also has a combinatorial system. There's a finite number of notes but a limitless number of musical compositions. DNA also has this sort of combinatorial system where you have a finite number of, I guess, bases or amino acids that could combine to a possible infinity of strings, of DNA strings. So, how does this happen? Well, the infinity mechanism, and many of you will be familiar with this from mathematics or computer science, is recursion. And there's a lot to be said about this but it could be pretty simply illustrated in language.

So, here's an example of a simple language. It's not — It's actually close to how linguists describe normal languages, but it's very simple. It has three nouns, "Fred," "Barney" and "Wilma," and two verbs, "thinks" and "likes." A very simple language. And one rule. And the way to read this rule is you make a sentence by taking a noun, any noun, putting a verb after it, and then following that verb with a noun. Now, when you do this, how many — And then so, for instance, you get the sentence "Fred likes Wilma." When you do this, how many possible sentences are there?

Let me just take a second. Okay. Any guesses? Eighteen. The sentences are "Fred likes Fred," "Fred likes Barney," "Fred likes Wilma," "Fred thinks Fred," "Fred thinks Barney," "Fred thinks Wilma," and

so on. The three nouns followed by any of the two verbs followed by any of the three nouns. That is not a very interesting language. But now, take a more complicated language — same vocabulary, the same three nouns, the same two verbs, the same sentence, but now one other sentence. This sentence expands to a noun followed by a verb followed by a sentence and there you get recursion. You have one rule invoking another rule and then you can get a sentence like "Fred thinks Barney likes Wilma." And here you get a potential infinity of sentences.

And this is obviously a toy example but you could see the use of recursion in everyday life and in everyday use of language. You could say, "John hates cheese," "My roommate heard a rumor that John hates cheese," "It disturbed Mary when I told her that my roommate heard a rumor that John hates cheese," "I was amazed that it disturbed Mary when I told her that my roommate heard a rumor that John hates cheese," "Professor Bloom had devoted way too much of his lecture talking about how I was amazed [laughter] that it disturbed Mary when I told her that my roommate heard a rumor that John hates cheese," "It really bothered me that — " and there's no limit. There's no longest sentence. You could keep producing a sentence deeper and deeper embedded until you die. And this is part of the power of language.

Now, the syntactic rules are complicated. And one of the puzzles of syntactic rules, or one of the issues of them, is that different rules can conspire to create the same sentence. So, you take a sentence like — This is a classic line from Groucho Marx: "I once shot an elephant in my pajamas. How it got into my pajamas I'll never know." And the humor, such that it is, revolves around the ambiguity of rules that generate it, like this versus like this. Often, to illustrate the issues of ambiguity, people have collected poorly thought-out headlines in newspaper reports that play on — that inadvertently have ambiguity. "Complaints about NBA referees growing ugly." So, that's the beauty of that structure. "Kids make nutritious snacks." "No one was injured in a blast which was attributed to the buildup of gas by one town official." Last summer I was in Seoul visiting the — visiting Korea University and the big headline there on the front page was "General arrested for fondling privates." [laughter]

Now, there actually is — The ambiguity is actually quite difficult to avoid in the construction and understanding of sentences. It's one of the ways in which it's often difficult to write clearly, and in fact, there's a whole sub-field of the law involving the use of linguistic theory to disambiguate sentences both in the Constitution, in legislation, as well as in some criminal cases.

And there was, several years ago, a very serious criminal case that rested on a sentence. And here's what happened. There were two brothers, one of them retarded, and they get into a robbery. And a police officer sees them and points the gun at them. And one of the brothers points a gun at the police officer. The police officer shouts for the brother, the non-retarded brother, to drop the gun. Actually, he said, "Give me the gun." The retarded brother shouted, "Let him have it," whereupon the brother shot and killed the police officer. Now, the brother who did the shooting was plainly a murderer. What about that brother who shouted, "Let him have it"? Well, it depends on what he — on how you interpret that sentence because the sentence is beautifully ambiguous. It could mean "shoot him, let him have it," or it could mean "give him the gun, let him have it." And in fact, the trial, which I think somebody could — If people out there know about this, please send me an e-mail. My understanding was he was found guilty but a lot to turn on the ambiguity of a sentence.

I want to shift now and talk about where does all this knowledge come from but I'll stop and answer any questions about the material so far.

Chapter 5. Question and Answer on Language Structure [00:35:21]

What are your questions? Yes.

Student: How does syntax differ from grammar or are they exactly the same?

Professor Paul Bloom: Syntax — The question is, "How is syntax different from grammar?" They're exactly the same. Syntax is a more technical term but it means the same thing as grammar. Yes.

Student: You said that every normal human being that's born uses at some point or another some kind of language. Aren't there people who weren't born within a culture and grew up and who never really spoke a language though they were physically normal?

Professor Paul Bloom: Yes. I'm glad you actually asked me about that because, as I said it, I realized it wasn't quite right. The point that was just raised here is I had said before that everybody who's neurologically normal comes to acquire and learn a language. But what about people who are neurologically normal but they don't have language around them? And in fact, there have been, historically, some cases of this. There's been, probably apocryphal, stories about children who are raised by wolves or by dogs. There are stories, horrible stories, some in the twentieth century, about children who are locked away by insane or evil parents and have never learned to speak. There are stories of deaf people who are within certain societies where nobody signs to them, and so they're what's known as linguistic isolates. And they themselves never learn to speak. And those cases are the dramatic exception and they do tell you something.

They tell you that it's not enough to have a brain for language. Somebody does have to use it with you. Interestingly, it doesn't have to be that many people. So, Susan Goldin-Meadow has studied deaf children that nobody signed to but what she studies is deaf children with deaf siblings and these children don't just sit there. They create their own language. It's not a full-blown language like American sign-language or langue des signes quebecoise but it's a language nonetheless, with words and syntax and phonology. It's an interesting question. Any other questions? Yes.

Student: Could it be argued that there are inherent limits to grammar?

Professor Paul Bloom: It's a good question. The question is, "Are there inherent limits in our abilities to come up with grammars?" And most linguists would argue "yes," that languages are highly constrained in how they do things. So, for instance, one example is there's no language in the world that ever constructs a question by switching the order of words around in a sentence. There's no language in the world that has a rule that says the fifth word has to be a verb. And linguists have all of these conditions they say, "no language in the world works this way." Now this is — ;So, these are constraints on grammar and they're really interesting because they tell us what's a humanly natural language versus what's not a humanly natural language. But notice, even if there is incredible constraints on grammars, still — we could still produce an infinite number of sentences. It's just like if you restrict me to only a subset of numbers, only the odd numbers, still there's an infinity of odd numbers. So, grammar can be restricted but still give rise to an infinity of possible sentences.

Chapter 6. Noam Chomsky and Language Acquisition [00:39:10]

Well, there's a radical claim about the origin of language associated with the guy who we met when we talked about behaviorism who wrote A Review of Verbal Behavior, the linguist Noam Chomsky. And Chomsky makes this radical claim. And this is that we shouldn't view language learning as learning at all. Instead, we should view it as something similar to growth. So he says,

No one would take seriously the proposal that a human organism learns through experience to have arms rather than wings, or that the basic structure of particular organs results from accidental experience. [Language] proves to be no less marvelous and intricate than these physical structures. Why, then, should we not study the acquisition of a cognitive structure like language more or less as we study some complex bodily organ?

So, you might learn to play baseball, you might learn about the American Civil War, but if Chomsky is right you didn't learn to speak English. Rather, what happened is you heard English and — but the capacity grew in your head and something a lot more similar to the development of arms or legs or a visual system.

Well, should we believe this? We know there has to be some effect of the environment shaping language, obviously, because in order to know English you have to have heard English, in order to know Dutch you have had to heard, to — had to have learned and heard Dutch. And in fact, languages differ in all the ways that we were talking about. Some languages like English has a — have a distinction between ‘l' and ‘r.' Other languages do not. For a language like English, that creature there is referred to with the morpheme "dog." That's a historical accident of English. In French it's chien and in Greek it's something else. And each of those 6,000 languages and people in the room who know another language would say, "Yeah, in Vietnamese it's this," "In Urdu it's this," "In Czech it's that."

Finally, there is syntax. So, English is what's known as a subject-verb-object language. That means if you want to convey the idea that Bill hit John, you would say, "Bill hit John." But not all languages work that way. In fact, the majority of languages, more languages, are actually subject-object-verb languages. So, you would say, if you wanted to convey that Bill was the hitter and John was hit, "Bill John hit." All of this has to be learned. And all of this has to be learned through exposure to language users.

On the other hand, there is considerable evidence that the development of these language skills, in some way, is similar to growth in the way that Chomsky suggests. So, here are some basic facts about language development. One is something which I had mentioned before. All normal children learn language. There can be specific impairments of language. Now, again, we spoke about them before when talking about the brain. Some of these impairments could be due to trauma, the aphasias. Trauma, a blow to the head, a stroke can rid you of your language. But, also, there are genetic disorders, some falling under the rubric of what's known as "specific language impairment," where children are born without the same ability as the rest of us to learn to speak. And these are interesting in many ways.

One reason that they're interesting is that they illustrate something about human language. It is not — It would not be unreasonable for you to think before listening to his lecture, "Look. All you need to have to learn a language is to be smart" or "All you need to have to learn a language is to want to

communicate" or "All you need to have to learn a language is to be a social person wanting to — having the ability to understand others and deal with others." But the cases of specific language impairments suggest that all of that is wrong, because there are children in this world right now who are plenty smart, who really want to communicate, and who are entirely social creatures but they can't learn language. And this suggests that the ability to learn language and understand language is to some extent separate from these other aspects of mental life.

Continuing on this theme, we also know that language is learnt without any sort of feedback or training. There are many Americans who believe that they need to teach their children language. And there's a huge industry with DVDs and flash cards and all sorts of things designed to teach your children language. And I think many parents believe that if they didn't persist in using these things their children would never learn to speak. But we know that that's not true. We know that this isn't true because there are communities where they don't speak to their kids. They don't speak to their kids because they don't believe it's important to speak to their kids. Some linguists would interview — Linguists would interview adults in these communities and say, "Why don't you speak to your babies?" And these adults would respond, "It'd be ridiculous to speak to a baby. The baby has nothing to say. You might as well just speak to your dog." And then the American linguist would say, "Yeah. We speak to our dogs." [laughter] Americans and Europeans speak to everything and everybody. Other cultures are more picky and they don't talk to their children until their children themselves are talking. This doesn't seem to make much of a difference in language learning.

Some studies have, motivated by Chomsky's work in expressed — sorry, motivated by Chomsky's critique of Skinner's Verbal Behavior, have asked even in — "What if we just looked at children within the United States? Don't these children get feedback?" And the answer is yes and no. So your average highly educated Western parent does give their children feedback — do give their children feedback based on what they say. But they don't typically give feedback based on the syntax or grammaticality of what they say.

The example given by Brown and Hanlon in the classic study in the 1970s is they did all of these studies looking at what children say and how parents responded, and it turns out parents respond not to the grammatical correctness but to the affect or cuteness or sociability of the utterance. So for instance, if a child says to his mother, "I loves you, Mommy," it's a very unusual parent who would say, "Oh, no. The verb agreement is mistaken. [laughter] You've added a redundant ‘s.' It's not appropriate." Similarly, if a child is to say, "I hate your guts, Mother," it's an unusual mother, "That's wonderful. There's a subject, verb, object. The whole thing's structurally fine." We respond to our kids like we respond to each other based on the message that's conveyed, not the grammaticality of the utterances. Children make grammatical mistakes all the time but then they go away and they go away without correction. So those are some basic facts.

Chapter 7. The Time Course of Language Acquisition [00:47:07]

What do we know about the time course of language? Well, early on children start off and they prefer the melody of their own language. These studies were done in France with four-day-old babies. And what they did was they used a sucking method. Remember, there's a limited number of things babies can do. One of the things they can do is suck, and these babies would suck on a pacifier to hear French.

And they would prefer to hear French than to hear Russian. And these investigators claimed this is because they had been exposed to French in the first four days of their lives. Reviewers, mostly from France, objected and said, "No. Maybe French just sounds better. Everybody's going to like French." So, they re-did the study in Russia. Russian kids sucked harder to hear Russian than they did to French.

And what they're listening to isn't the words. They don't know words yet. They don't know of syntax yet. It's the rhythm of the language. For you, French and Russian sound different. Even if you're like me and you don't know a word of either language, they still sound different. They sound different to babies too. And a baby being raised in France or a baby being raised in Russia knows enough to tell what's his language and what isn't.

Early on, children are sensitive to every phoneme there is. So, English-speaking children, for instance, can — English-speaking babies – babies who are born in the United States – can distinguish between English phonemes like "lip" and "rip" but they could also distinguish between phonemic contrasts that are not exemplified in English, such as phonemic contrasts in Czech or Hindi. Yes.

Student: I'm wondering if you can say the wrong things to them — to infants based on what you were saying before. Because I was in France one summer and I had some neighbors there. I hated these neighbors, I thought they were stupid. Not because they were French, but they had a baby and it would gaggle and coo and they would respond in similar terms.

Professor Paul Bloom: They would gaggle and coo back at the baby.

Student: [inaudible] And I hate these people. [inaudible] So I don't know if it — Does it matter what you say to babies as long as you say something.

Professor Paul Bloom: There's a lot going on in your question. [laughter] Some raising — Well, there's a lot going on in your question. The answer to the question — The question was, "your baby's going to coo and 'ga ga, goo goo,' does it matter if you coo and 'ga ga, goo goo' back?" No, it doesn't make a difference. Your hatred towards them was unmotivated. You can be relieved of that debt, or now you know you feel bad now, I guess. [laughter] If you speak to your children in perfect English, it's very strange. Nobody speaks to their babies in, "Hello, Son. It's time — Oh. You want to change your diaper right now so stay still." That's bad parenting. It sounds kind of silly. More — What most people do is, "Oh. You're such a cute little baby." And it probably — One — There's — Evolutionary psychologists debate the function of why we talk funny to babies. And some people have argued that it does help their language learning. And some people have argued instead that what it does is it calms them. They like to hear the music of a smooth voice and so on. But whether or not you do so doesn't seem to make a big difference.

It is very difficult to find any effect of how parents talk to their kids on how their kids learn language, particularly when it comes to babies. So, early on babies can — are sensitive to all phonemes and then that goes away. Around twelve months of age it goes away. This is one thing you were much better at when you were a baby than you are now. When you were a baby you were a multilingual fool. You could understand the sound differences of every language on earth. Now, if you're like me, you could barely understand English. [laughter] You narrow down until you're sensitive just to the language you hear. And this narrowing down is largely in place by about twelve months of age.

Around seven months is babbling. And I want to stop at this point to go back to the issue — I promised you I would turn a bit to sign language and I want to describe now a very elegant — I want to show a little film now of a very elegant series of experiments looking at the question of whether babies who are exposed to a sign language, babble.

[film plays]

One of the real surprising findings in my field over the last ten/twenty years has been that the acquisition of sign languages has turned out to be almost exactly the same; in fact, as far as we know, exactly the same as the acquisition of spoken languages. It didn't have to be that way. It could have been just as reasonable to expect that there'd be an advantage for speech over sign. That sign languages may be full-blown languages but they just take — they're just harder to learn because the brain and the body have adapted for speech. It turns out that this just isn't the case. It turns out that sign and — the developmental milestones of sign languages and the developmental milestones of spoken languages are precisely the same. They start babbling at the same point. They start using first words, first sentences, first complicated constructions. There seems to be no interesting difference between how the brain comes to acquire and use the spoken language versus a sign language.

Around twelve months of age, children start using their first words. These are words for objects and actions like "dog" and "up" and "milk." They start showing some sensitivity to the order of words. So they know that "dog bites cat" is different from "cat bites dog." Around eighteen months of age, they start learning words faster. They start producing little, miniature sentences like "Want cookie" or "Milk spill" and the function morphemes, the little words, "in," "of," "a," "the," and so on start to gradually appear.

Then the — Then there's the bad news. Around seven years of age going up through puberty, the ability to learn language starts to go away. The best work on this has been done by Elissa Newport and Sam Supalla who have studied people who have been in the United States for many, many years – 30, 40 years – and seeing how well they have come to speak English. And it turns out the big determinant of how well you speak English as an immigrant isn't how smart you are. It's not how many family members you have when you're here. It's not your motivation. It's how old you were when you started.

It turns out that if you start learning a language – a second language is where most of the work's been done – within the first few years of life you're fine. You'll speak like a native. But then it starts getting worse and worse. And once you hit puberty, suddenly there's huge variation in the abilities you have to learn language. It is very rare, for instance, for somebody who has learned English past puberty to speak without an accent. An accent is very hard to shake and it's not just an accent. It's also other aspects of phonology, syntax, and morphology. It's like the part of the brain that's responsible for language learning is only around early in development and if you don't get your language by then it'll just run out.

I want to begin next class with this question, the question of animals. And that will shut down the language learning part. But one thing I'll put up here is your second reading response. So, I'll also put this up on Wednesday, and by Wednesday you might have a bit of a better — be in a better position to answer this question. But I'll continue with language on Wednesday and then we'll also talk about vision, attention, and memory. I'll see you then.